Ordinary Differential Equations

Lecture Notes

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Lecture 8: Applications in Science and Engineering

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1 Introduction to ODE Applications

Ordinary differential equations serve as the mathematical foundation for modeling dynamic processes across virtually every field of science and engineering. From the microscopic behavior of molecules to the macroscopic evolution of galaxies, from the spread of diseases to the dynamics of financial markets, ODEs provide the language for describing how systems change over time. This universality stems from the fundamental nature of differential equations as mathematical expressions of physical laws, conservation principles, and empirical relationships.

The power of ODE modeling lies not merely in describing observed phenomena but in predicting future behavior, understanding system sensitivity to parameters, and designing interventions to achieve desired outcomes. Modern computational capabilities have dramatically expanded the scope of problems that can be addressed, enabling the study of complex, multi-scale systems that were previously intractable.

This lecture explores representative applications that demonstrate both the breadth of ODE applications and the depth of insight they provide. We examine mechanical systems that exhibit the full spectrum of dynamical behavior, biological systems that reveal the complexity of living processes, electrical circuits that form the basis of modern technology, and chemical reactions that drive both industrial processes and biological function.

Each application area presents unique modeling challenges and opportunities. Mechanical systems often involve conservation laws and geometric constraints that must be respected in both analytical and numerical treatments. Biological systems typically exhibit nonlinear interactions, multiple time scales, and stochastic effects that require sophisticated modeling approaches. Electrical circuits combine linear and nonlinear elements in networks that can exhibit complex dynamics. Chemical systems involve mass action kinetics and thermodynamic constraints that shape their temporal evolution.

1.1 Modeling Workflow and Principles

Effective mathematical modeling requires a systematic approach that balances physical realism with mathematical tractability. The modeling process typically involves several key steps that transform real-world phenomena into mathematical frameworks amenable to analysis and computation.

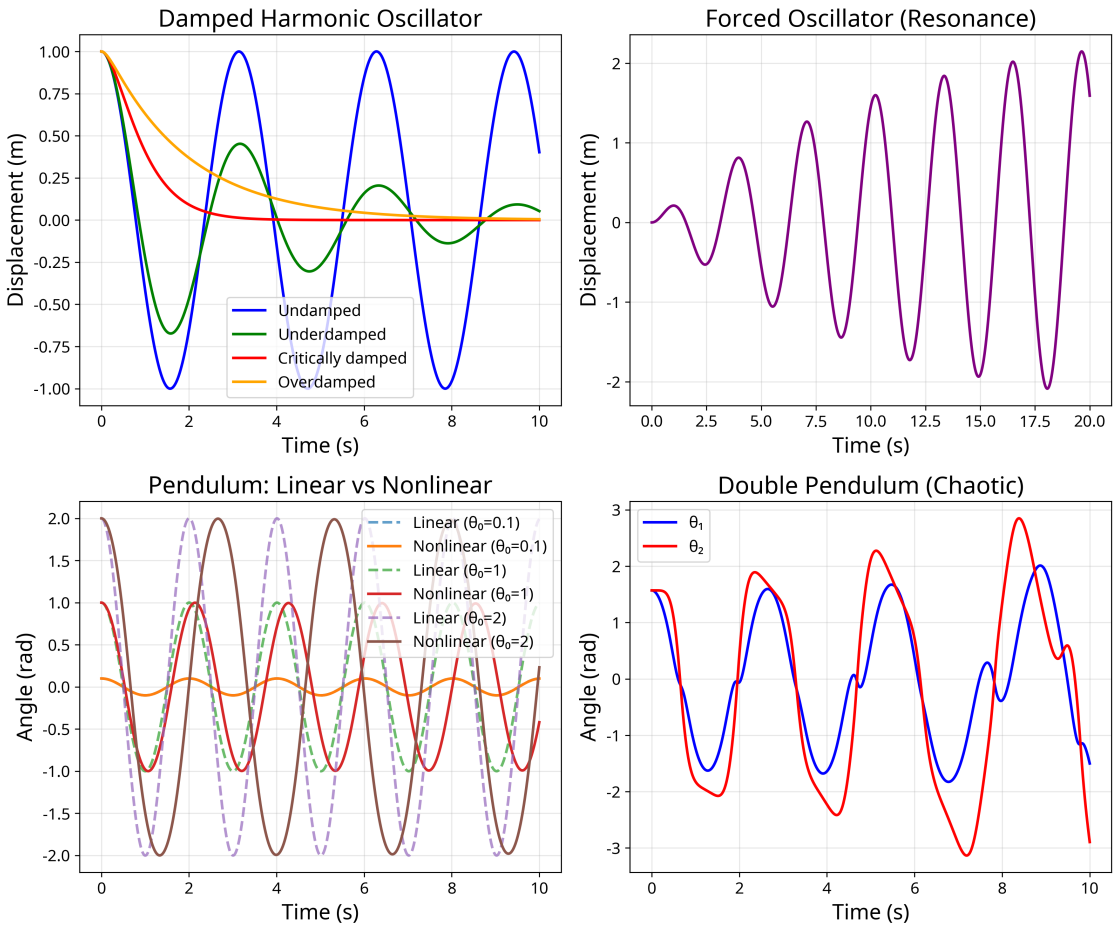

Figure 1: Mechanical system applications: damped harmonic oscillator with different damping regimes, forced oscillator showing resonance, linear vs nonlinear pendulum comparison, and chaotic double pendulum dynamics.

Problem Identification and Scope Definition: The first step involves clearly defining the system of interest, identifying the key variables and parameters, and establishing the temporal and spatial scales relevant to the problem. This scoping process determines which physical effects must be included and which can be neglected or approximated.

Physical Principle Identification: Most ODE models derive from fundamental physical principles such as conservation of mass, energy, and momentum, Newton's laws of motion, Kirchhoff's laws for electrical circuits, or empirical relationships like Fick's law for diffusion. Identifying the relevant principles provides the foundation for mathematical formulation.

Mathematical Formulation: The physical principles are translated into mathematical equations involving derivatives of the system variables. This step often requires making simplifying assumptions about system geometry, material properties, or interaction mechanisms.

Dimensionless Analysis: Converting equations to dimensionless form reveals the fundamental parameter groups that control system behavior and enables the identification of different dynamical regimes. This analysis often provides crucial insights into system scaling and parameter sensitivity.

Model Validation and Refinement: Comparing model predictions with experimental data or known analytical solutions validates the model and identifies areas where refinement may be

needed. This iterative process gradually improves model fidelity and predictive capability.

The applications examined in this lecture illustrate these principles while demonstrating the rich variety of phenomena that can be captured by ODE models.

2 Biological Systems

Biological systems present unique modeling challenges due to their complexity, nonlinearity, and multi-scale nature. ODE models have proven invaluable for understanding population dynamics, epidemiology, biochemical networks, and physiological processes.

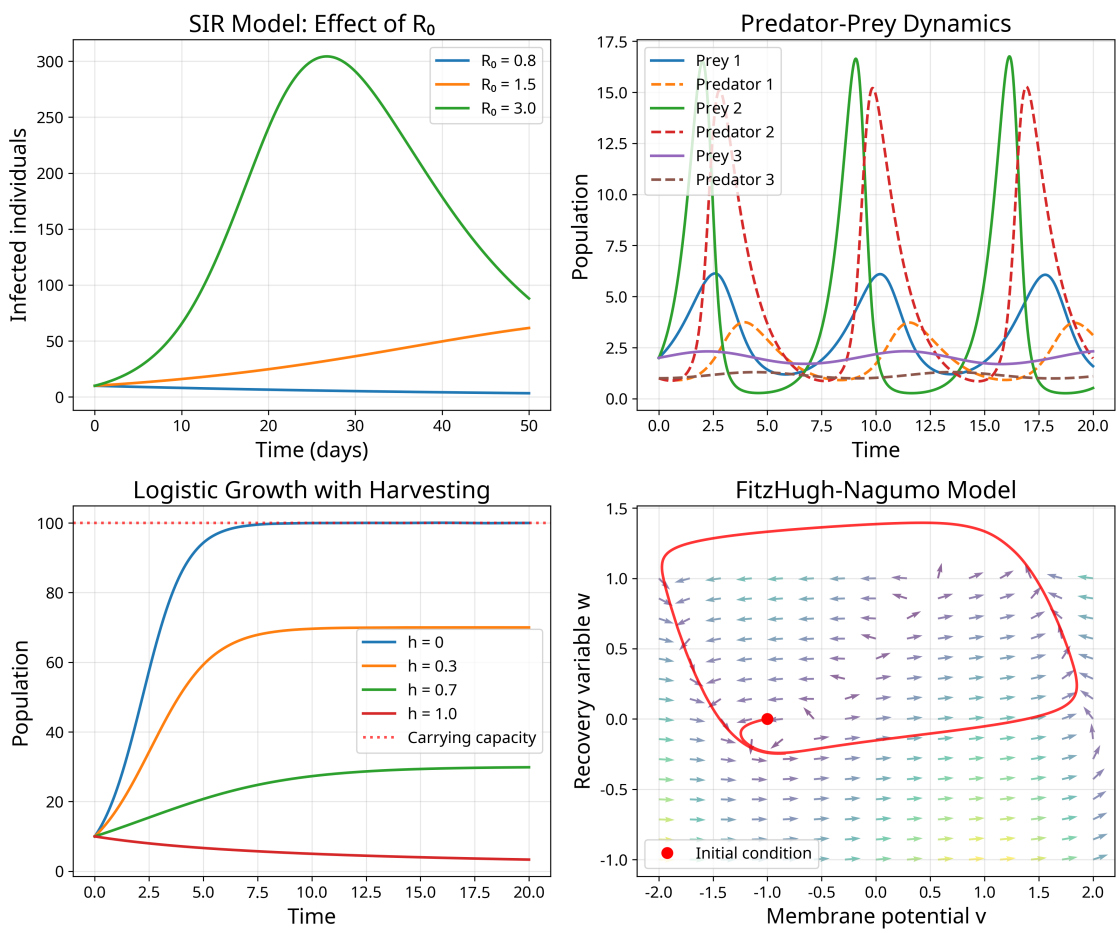

Figure 2: Biological applications: SIR epidemic model with varying reproduction numbers, predator-prey dynamics with different parameters, logistic growth with harvesting effects, and FitzHugh-Nagumo neural excitation model.

2.1 Population Dynamics

Population models form the foundation of ecology, conservation biology, and resource management. These models capture the essential dynamics of birth, death, and interaction processes that determine population changes over time.

Exponential and Logistic Growth: The simplest population model assumes exponential growth:

$$\frac{dN}{dt} = rN \tag{1}$$

where $N(t)$ is population size and r is the intrinsic growth rate. This gives unlimited exponential growth $N(t) = N_0 e^{rt}$, which is unrealistic for finite environments.

The logistic model incorporates carrying capacity K:

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right) \tag{2}$$

The solution approaches the carrying capacity sigmoidally:

$$N(t) = \frac{K}{1 + \left(\frac{K}{N_0} - 1\right) e^{-rt}} \tag{3}$$

The logistic model exhibits a single stable equilibrium at $N = K$, representing the balance between growth potential and environmental limitations.

Predator-Prey Dynamics: The Lotka-Volterra model describes interacting predator and prey populations:

$$\frac{dx}{dt} = ax - bxy \tag{4}$$

$$\frac{dy}{dt} = -cy + dxy \tag{5}$$

where x is prey density, y is predator density, and $a, b, c, d > 0$ are rate parameters. This system conserves the quantity:

$$H(x, y) = dx + by - c\ln x - a\ln y \tag{6}$$

The phase portrait consists of closed orbits around the equilibrium $(c/d, a/b)$, representing periodic oscillations in both populations. The predator population lags behind the prey population, creating the characteristic phase relationship observed in many natural systems.

More realistic models include carrying capacity for prey, predator saturation effects, and additional mortality terms:

$$\frac{dx}{dt} = rx\left(1 - \frac{x}{K}\right) - \frac{axy}{1 + hx} \tag{7}$$

$$\frac{dy}{dt} = \frac{eaxy}{1 + hx} - my \tag{8}$$

These modifications can lead to stable equilibria, limit cycles, or more complex dynamics depending on parameter values.

Example. Consider a fish population subject to harvesting at rate H:

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right) - H \tag{9}$$

For constant harvesting, equilibria occur where growth equals harvest rate. The maximum sustainable yield occurs at $N = K/2$, giving $H_{\max} = rK/4$.

If $H > H_{\max}$, no equilibrium exists and the population crashes to extinction. This demonstrates the critical importance of harvest rate control in sustainable resource management.

More sophisticated models include age structure, spatial distribution, and economic factors, but the basic principle of balancing growth and harvest remains fundamental to fisheries science.

2.2 Epidemiological Models

Mathematical epidemiology uses ODE models to understand disease spread and evaluate intervention strategies. These models have become increasingly important for public health planning and policy development.

SIR Model: The basic SIR (Susceptible-Infected-Recovered) model divides the population into three compartments:

$$\frac{dS}{dt} = -\beta SI \tag{10}$$
$$\frac{dI}{dt} = \beta SI - \gamma I \tag{11}$$
$$\frac{dR}{dt} = \gamma I \tag{12}$$

where $S + I + R = N$ (constant total population), β is the transmission rate, and γ is the recovery rate.

The basic reproduction number $R_0 = \beta N/\gamma$ determines epidemic behavior:

- If $R_0 < 1$, the disease dies out
- If $R_0 > 1$, an epidemic occurs

The final epidemic size satisfies the transcendental equation:

$$S_\infty = S_0 e^{-R_0(1-S_\infty/N)} \tag{13}$$

This relationship shows that not everyone becomes infected even in a severe epidemic, as the depletion of susceptibles eventually stops transmission.

SEIR Model: Adding an exposed (latent) class accounts for incubation periods:

$$\frac{dS}{dt} = -\beta SI \tag{14}$$
$$\frac{dE}{dt} = \beta SI - \sigma E \tag{15}$$
$$\frac{dI}{dt} = \sigma E - \gamma I \tag{16}$$
$$\frac{dR}{dt} = \gamma I \tag{17}$$

The exposed class represents individuals who are infected but not yet infectious. This model better captures diseases with significant incubation periods like COVID-19, influenza, or measles.

Vaccination and Control: Vaccination can be incorporated by modifying the susceptible equation:

$$\frac{dS}{dt} = -\beta SI - \nu S \tag{18}$$

where ν is the vaccination rate. The critical vaccination coverage needed to prevent epidemics is:

$$p_c = 1 - \frac{1}{R_0} \tag{19}$$

This herd immunity threshold shows that not everyone needs to be vaccinated to prevent disease spread, but the required coverage increases with disease transmissibility.

2.3 Biochemical Networks

Cellular processes involve complex networks of biochemical reactions that can be modeled using systems of ODEs based on mass action kinetics and enzyme kinetics.

Enzyme Kinetics: The Michaelis-Menten mechanism describes enzyme-catalyzed reactions:

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} E + P \tag{20}$$

The full system of ODEs is:

$$\frac{d[S]}{dt} = -k_1[E][S] + k_{-1}[ES] \tag{21}$$

$$\frac{d[ES]}{dt} = k_1[E][S] - k_{-1}[ES] - k_2[ES] \tag{22}$$

$$\frac{d[P]}{dt} = k_2[ES] \tag{23}$$

with conservation laws $[E]_0 = [E] + [ES]$ and $[S]_0 = [S] + [ES] + [P]$.

Under the quasi-steady-state approximation ($\frac{d[ES]}{dt} \approx 0$), this reduces to the Michaelis-Menten equation:

$$\frac{d[P]}{dt} = \frac{V_{\max}[S]}{K_M + [S]} \tag{24}$$

where $V_{\max} = k_2[E]_0$ and $K_M = (k_{-1} + k_2)/k_1$.

Gene Regulatory Networks: Gene expression can be modeled using Hill functions to capture cooperative binding:

$$\frac{dx}{dt} = \frac{\alpha}{1 + (y/K)^n} - \delta x \tag{25}$$

where x is the protein concentration, y is a repressor concentration, n is the Hill coefficient (cooperativity), and α, K, δ are kinetic parameters.

Networks of such equations can exhibit bistability, oscillations, and other complex behaviors essential for cellular function. The lac operon, circadian clocks, and cell cycle control all involve regulatory circuits that can be analyzed using ODE models.

3 Chemical Reaction Systems

Chemical kinetics provides another rich source of ODE applications, with reactions governed by mass action laws and conservation principles.

3.1 Elementary Reaction Kinetics

Mass Action Law: For an elementary reaction $aA + bB \to cC + dD$, the reaction rate is:

$$r = k[A]^a[B]^b \tag{26}$$

where k is the rate constant and $[X]$ denotes the concentration of species X.

First-Order Reactions: The simple decay $A \to B$ gives:

$$\frac{d[A]}{dt} = -k[A] \tag{27}$$

$$\frac{d[B]}{dt} = k[A] \tag{28}$$

with solution $[A](t) = [A]_0 e^{-kt}$ and $[B](t) = [A]_0(1 - e^{-kt})$.

Second-Order Reactions: For $A + B \to C$:

$$\frac{d[A]}{dt} = -k[A][B] \tag{29}$$

$$\frac{d[B]}{dt} = -k[A][B] \tag{30}$$

$$\frac{d[C]}{dt} = k[A][B] \tag{31}$$

If $[A]_0 = [B]_0 = a$, then $[A](t) = [B](t) = \frac{a}{1+akt}$.

3.2 Complex Reaction Networks

Real chemical systems involve networks of coupled reactions that can exhibit complex dynamics.

Consecutive Reactions: For the sequence $A \xrightarrow{k_1} B \xrightarrow{k_2} C$:

$$\frac{d[A]}{dt} = -k_1[A] \tag{32}$$

$$\frac{d[B]}{dt} = k_1[A] - k_2[B] \tag{33}$$

$$\frac{d[C]}{dt} = k_2[B] \tag{34}$$

The intermediate B exhibits a maximum concentration at time $t_{\max} = \frac{\ln(k_2/k_1)}{k_2 - k_1}$ (for $k_2 \neq k_1$).

Autocatalytic Reactions: Reactions where a product catalyzes its own formation can exhibit sigmoidal growth:

$$A + B \to 2B \tag{35}$$

gives:

$$\frac{d[A]}{dt} = -k[A][B] \tag{36}$$

$$\frac{d[B]}{dt} = k[A][B] \tag{37}$$

With conservation $[A] + [B] = [A]_0 + [B]_0$, this becomes logistic growth for $[B]$ (see 'Biological Systems → Population Dynamics').

Oscillating Reactions: The Brusselator model demonstrates how chemical reactions can produce sustained oscillations:

$$A \to X \tag{38}$$

$$2X + Y \to 3X \tag{39}$$

$$B + X \to Y + D \tag{40}$$

$$X \to E \tag{41}$$

Assuming constant concentrations of A and B, the rate equations are:

$$\frac{d[X]}{dt} = A - (B+1)[X] + [X]^2[Y] \tag{42}$$

$$\frac{d[Y]}{dt} = B[X] - [X]^2[Y] \tag{43}$$

For appropriate parameter values, this system exhibits limit cycle oscillations, demonstrating that chemical systems can maintain periodic behavior far from equilibrium.

4 Multi-Scale and Coupled Systems

Many real-world applications involve multiple time scales, spatial scales, or coupled subsystems that require sophisticated modeling approaches.

4.1 Stiff Systems and Multiple Time Scales

Systems with widely separated time scales pose significant challenges for both analytical and numerical treatment.

Singular Perturbation Methods: For systems of the form:

$$\frac{dx}{dt} = f(x, y, \epsilon) \tag{44}$$

$$\epsilon \frac{dy}{dt} = g(x, y, \epsilon) \tag{45}$$

where $0 < \epsilon \ll 1$, the variable y evolves much faster than x. Singular perturbation theory provides systematic methods for analyzing such systems by identifying fast and slow manifolds.

Quasi-Steady-State Approximation: When variables equilibrate rapidly, a QSSA reduces system dimension and stiffness; for a concrete example, see 'Biological Systems → Biochemical Networks → Enzyme Kinetics (Michaelis–Menten)'.

4.2 Coupled Oscillator Systems

Networks of coupled oscillators appear throughout science and engineering, from mechanical systems to biological rhythms to power grids.

Kuramoto Model: A paradigmatic model for synchronization in oscillator networks:

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^{N} \sin(\theta_j - \theta_i) \tag{46}$$

where θ_i is the phase of oscillator i, ω_i is its natural frequency, and K is the coupling strength.

For weak coupling, oscillators remain incoherent. Above a critical coupling strength, partial synchronization emerges, with the order parameter:

$$re^{i\psi} = \frac{1}{N} \sum_{j=1}^{N} e^{i\theta_j} \tag{47}$$

measuring the degree of synchronization.

Mechanical Coupled Oscillators: Two masses connected by springs exhibit normal modes:

$$m_1 \frac{d^2 x_1}{dt^2} = -k_1 x_1 + k_c(x_2 - x_1) \tag{48}$$

$$m_2 \frac{d^2 x_2}{dt^2} = -k_2 x_2 - k_c(x_2 - x_1) \tag{49}$$

The normal mode frequencies are determined by the eigenvalues of the system matrix, and general motion is a superposition of these modes.

5 Modern Applications and Emerging Areas

Contemporary applications of ODEs continue to expand into new domains driven by technological advances and interdisciplinary research.

5.1 Systems Biology and Synthetic Biology

Modern molecular biology increasingly relies on quantitative models to understand cellular processes and design synthetic biological systems.

Circadian Rhythms: Biological clocks involve transcriptional-translational feedback loops that can be modeled as coupled oscillators:

$$\frac{dm}{dt} = \alpha_m \frac{K_I^n}{K_I^n + P^n} - \beta_m m \tag{50}$$

$$\frac{dP}{dt} = \alpha_p m - \beta_p P \tag{51}$$

where m is mRNA concentration and P is protein concentration. The Hill function captures the repressive effect of the protein on its own transcription.

Synthetic Gene Circuits: Engineered biological systems use ODE models for design and optimization. Toggle switches, oscillators, and logic gates can all be implemented using genetic regulatory circuits with predictable dynamics.

5.2 Climate and Environmental Modeling

Climate systems involve coupled atmosphere-ocean-land dynamics that can be studied using ODE models for key processes.

Energy Balance Models: Simple climate models treat Earth's temperature as governed by:

$$C\frac{dT}{dt} = S(1-\alpha) - \sigma T^4 + \Delta F \tag{52}$$

where C is heat capacity, S is solar constant, α is albedo, σ is the Stefan-Boltzmann constant, and ΔF represents radiative forcing from greenhouse gases.

Carbon Cycle Models: The global carbon cycle can be modeled as a network of reservoirs (atmosphere, ocean, biosphere) connected by fluxes governed by ODE systems.

5.3 Financial Mathematics

Financial markets exhibit complex dynamics that can be modeled using stochastic differential equations and deterministic ODE models.

Option Pricing: The Black-Scholes equation for option pricing is a parabolic PDE that can be solved using ODE methods after appropriate transformations.

Market Dynamics: Models of market behavior often involve coupled equations for price, volume, and volatility that exhibit complex dynamics including bubbles, crashes, and regime changes.

Exercises

Exercise 1

A damped, forced oscillator (e.g., a base-isolated structure) obeys

$$mx'' + cx' + kx = F_0 \cos(\omega t).$$

Non-dimensionalize using $t = \tau/\omega_0$ with $\omega_0 = \sqrt{k/m}$ and $x = Xx_0$ to obtain

$$x_{\tau\tau} + 2\zeta\, x_\tau + x = \Gamma \cos(\Omega\tau).$$

Identify $\zeta,\ \Omega,\ \Gamma$ and explain the modeling meaning of each parameter.

Exercise 2

A population with constant harvesting satisfies

$$N' = rN\Big(1 - \tfrac{N}{K}\Big) - H, \qquad r > 0,\ K > 0.$$

Find all equilibria as functions of H, determine their stability, and compute the maximum sustainable harvest $H_{\max}$.

Exercise 3

An SIR epidemic model is given by

$$S' = -\beta SI, \quad I' = \beta SI - \gamma I, \quad R' = \gamma I, \quad N = S + I + R.$$

Compute $R_0 = \beta N/\gamma$ and the critical vaccination coverage $p_c = 1 - 1/R_0$. Use

$$S_\infty = S_0 \exp\big(-R_0(1 - S_\infty/N)\big)$$

to determine S_∞ numerically for specified $N, \beta, \gamma, S_0, I_0$.

Exercise 4

Starting from the enzyme mechanism

$$E + S \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} ES \xrightarrow{k_2} E + P,$$

write the ODEs with the conservation $[E]_0 = [E] + [ES]$. Using the quasi–steady-state assumption for $[ES]$, derive

$$v = \frac{d[P]}{dt} = \frac{V_{\max}[S]}{K_M + [S]}, \qquad V_{\max} = k_2[E]_0, \quad K_M = \frac{k_{-1} + k_2}{k_1}.$$

Exercise 5

A self-activating gene can be modeled by

$$\frac{dx}{dt} = \frac{\alpha}{1 + (x/K)^n} - \delta x, \qquad \alpha, \delta, K > 0,\ n \geq 1.$$

Find equilibria in terms of (α, δ, K, n). Show uniqueness for $n = 1$. For $n > 1$, discuss when multiple equilibria can occur by inspecting the nullcline shape, and classify stability using $f'(x^*)$.

Exercise 6

Two weakly coupled phase oscillators satisfy

$$\dot{\theta}_1 = \omega_1 + \tfrac{K}{2}\sin(\theta_2 - \theta_1), \qquad \dot{\theta}_2 = \omega_2 + \tfrac{K}{2}\sin(\theta_1 - \theta_2).$$

Derive the phase-difference equation $\dot{\phi} = \Delta - K\sin\phi$ with $\phi = \theta_1 - \theta_2$ and $\Delta = \omega_1 - \omega_2$. Find the locking condition on K and the stable locked phase ϕ^* when $|\Delta| \leq K$.

Exercise 7

The Brusselator chemical oscillator is modeled by

$$x' = A - (B+1)x + x^2 y, \qquad y' = Bx - x^2 y.$$

Find the equilibrium (x^*, y^*), compute the Jacobian there, and give conditions on (A, B) for linear stability. For $A = 1$, locate the Hopf threshold in B and decide stability below and above that value.

Exercise 8

A zero-dimensional climate energy-balance model reads

$$C\,\dot{T} = S(1-\alpha) - \sigma T^4 + \Delta F, \qquad \bar{T} = 288\,\mathrm{K}.$$

Linearize about $\bar{T}$ to obtain $C\,\dot{\theta} = -\lambda\theta + \Delta F$ with $\theta = T - \bar{T}$ and $\lambda = 4\sigma\bar{T}^3$. Compute the relaxation time $\tau = C/\lambda$ and interpret it physically.